

Daniele Friolo



Contact

Daniele Friolo
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Languages

Fluent English and
Italian

Programming Languages

C, Python, Java, SQL,
Solidity.

Education

2021	Ph.D. in Cryptography Thesis: New Perspectives in Multi-Party Computation: Low Round Complexity from New Assumptions, Financial Fairness and Public Verifiability Supervisor: Prof. Daniele Venturi	Sapienza University of Rome - CS Dept.
2017	Master Degree in Computer Science Thesis: Predictable Arguments Advisor: Prof. Daniele Venturi	Sapienza University of Rome - CS Dept.
2015	Bachelor Degree in Computer Science Thesis: Android Client and Soa Server Mobile App for Car Pooling Advisor: Prof. Andrea Sterbini	Sapienza University of Rome - CS Dept.

Academia

April 2023 – March 2026	Assistant Professor (RTD-A)	Sapienza University of Rome. CS Dept.
2021–2023	Postdoctoral Researcher Supervisor: Prof. Daniele Venturi	Sapienza University of Rome. CS Dept.
2020–2021	Research Fellow Supervisor: Prof. Ivan Visconti	DIEM - University of Salerno.

Research Interests

I have been an Assistant Professor (RTD-A) at the Computer Science Department of Sapienza University of Rome under the PNRR SERICS Grant, where I am a Work-Package Leader for the Spoke-9 project SmartDeFi. I work on cryptography. My research topics vary from **Post-Quantum Cryptography, Secure Multi-Party Computation, Advanced Encryption Schemes, Zero-Knowledge, Subversion-Resilient Cryptography, and Blockchain Applications**. As a research fellow, I worked on Privacy and Cryptography on Blockchains with the research group of Prof. Ivan Visconti under the PRIViLEDGE project, Horizon 2020. In 2021, I was a Postdoctoral Researcher in Distributed Protocols for Digital Contact Tracing during the pandemic under the supervision of Prof. Daniele Venturi. During the third year of my Ph.D. I had the chance to visit and work with the Aarhus Crypto Group in Denmark, hosted by Prof. Ivan Damgård, and given an invited talk at the University of Lund and Chalmers University in Sweden. In 2023, I had the chance to work for 6 months as a visiting scholar with Prof. Giuseppe Ateniese at George Mason University in Virginia, USA. I am currently part of the Sapienza's cryptography research group SCRYPTO Lab (<https://scrypto.di.uniroma1.it/>), led by Prof. Daniele Venturi.

I've published 17 papers in various venues, including top-notch IACR-recognised cryptography conferences such as Asiacrypt, Eurocrypt, and TCC.

I am workshop chair of the top-tier conference EUROCRYPT 2026, in Rome.

Research Visits

2025	Visiting Researcher - 1 week Hosted by Prof. Luisa Siniscalchi, I worked on anamorphic encryption-related projects.	Technical University of Denmark, Denmark
2023	Visiting Researcher - 6 months Hosted by Prof. Giuseppe Ateniese, I worked on blockchain and cryptocurrency-related projects.	George Mason University (USA)
2019	Visiting Researcher - 1 year Hosted by Prof. Ivan Damgård, I worked together with Aarhus Crypto Group on Multi-Party Computation.	Aarhus University (DK)

Research Activities

Decentralized Finance: As a Work-Package Leader of the Project SmartDeFi, funded by PNRR, in coordination with the partners of the University of Genova (UNIGE) and the University of Milan (UNIMI), I produced the project deliverables for the Work Package 1 for Resilient Smart Contract Development, Cryptographic Tools, and Design of Large Scale Market Exchange. As a member of the SmartDeFi project, I developed new cryptographic primitives for Advanced Encryption Schemes [7,8,11], Enhanced Zero-knowledge proofs [4,9], protocols for efficient blockchain-based Central Bank Digital Currencies [16] and anonymous identity-based payments [17].

Privacy in Blockchains As a member of the PRIViLEDGE project, funded by European Union's Horizon 2020 research and innovation programme under grant agreement, I conducted research on blockchain-based attacks against digital contact tracing systems and developed a Toolkit allowing MPC protocol execution with a blockchain as a communication channel with reduced communication time (<https://github.com/danielefriolo/ledgerMPC>).

Multi-Party Computation During my PhD, I conducted research on fully simulatable oblivious transfer protocols [1], financially and cryptographically fair MPC protocols [6], and fork-resilient blockchain-based protocols [2].

Advanced Encryption Schemes: During my post-doc, I developed Advanced Encryption schemes such as Multi-Input Predicate Encryption [7,8] and Registered Functional Encryption [11] allowing for decryption against multiple ciphertexts and removal of trusted public key infrastructure in Functional Encryption schemes.

Post-Quantum Cryptography: In most of the conducted research, constructing post-quantum secure primitives is one of my main research objectives. Indeed, many of the developed works can be instantiated from standardized post-quantum cryptographic assumptions such as LWE and SIS.

Selected Publications

1. **A Black-Box Construction of Fully-Simulatable, Round-Optimal Oblivious Transfer from Strongly Uniform Key Agreement**, Friolo D., Masny D., Venturi D., in Proceedings of Theory of Cryptography Conference (TCC) 2019, Nuremberg, Germany.
2. **A Secure, Confidential, and Verifiable Decision Support System**, Edoardo Marangone, Eugenio Nemmi, Daniele Friolo, Giuseppe Ateniese, Ingo Weber, Claudio Di Ciccio, in Proceedings of the 31th European Symposium on Research in Computer Security (ESORICS) 2026, Rome, Italy.
3. **Shielded Computations in Smart Contracts Overcoming Forks**, Botta V., Friolo D., Visconti I., Venturi D., in Proceedings of Financial Cryptography and Data Security 2021, Virtual.
4. **Stealth and beyond: attribute-driven accountability in Bitcoin transactions**, Mongardini A. M., Friolo D. and Ateniese G. in Proceeding of the 24th International Conference on Applied Cryptography and Network Security (ACNS 2026), Stonybrook, NY, USA.

5. **Terrorist Attacks for Fake Exposure Notifications in Contact Tracing System**, Avitabile G., Friolo D., Visconti I., in Proceedings of the 19th International Conference on Applied Cryptography and Network Security 2021, Virtual
6. **Efficient Proofs of Knowledge for Threshold Relations**, Avitabile G., Botta V., Friolo D., Visconti I., in Proceedings of the 27th European Symposium on Research in Computer Security (ESORICS) 2022, Copenhagen, Denmark.
7. **Cryptographic and Financial Fairness**, Friolo D., Nam Ngo C., Massacci F., Venturi D. in IEEE Transactions on Information Forensics and Security.
8. **Multi-Key and Multi-Input Predicate Encryption from Learning with Errors**, Francati D., Friolo D., Malavolta G., Venturi D. in Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT) 2022
9. **Multi-Key and Multi-Input Predicate Encryption (for Conjunctions) from Learning with Errors**, Francati D., Friolo D., Malavolta G., Venturi D. in Journal of Cryptology.
10. **Self-Guarding Arbitrary Cryptographic Primitives and 2PC Protocols**, Friolo D., Reale A., Venturi D., Weber I., in CSF 2026 – 39th IEEE Computer Security Foundations Symposium (to appear).
11. **Registered (Inner-Product) Functional Encryption**, Francati, D., Friolo, D., Maitra, M., Malavolta, G., Rahimi, A., Venturi, D. in International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT) 2023, Guangzhou, China
12. **Data Redaction in Smart-Contract-Enabled Blockchains**, Avitabile, A., Botta V., Friolo, D., Visconti, I. in Blockchain: Research And Applications Journal.

Other Publications

13. **Affordable Security or Big Guy vs Small Guy**, Friolo D., Nam Ngo C., Massacci F., Venturi D. , in Security Protocols Workshop 2019
14. **Vision: What If They All Die? Crypto Requirements For Key People**, Ngo C., Friolo D., Massacci F., Venturi D., Battaiola E., in EuroUSEC 2020 Workshop, Virtual.
15. **Data Redaction in Smart-Contract-Enabled Blockchains**, Avitabile, A., Botta V., Friolo, D., Visconti, I. in 6th Distributed Ledger Technologies Workshop (DLT) 2024, Turin, Italy
16. **Private Electronic Payments with Self-Custody and Zero-Knowledge Verified Reissuance**, Friolo, D., Goodell G., Toliver D. and Al Nakib H. D. in Financial Cryptography Workshops 2025, Miyakojima, Japan.
17. **Stealth and Beyond: Attribute-Driven Accountability in Bitcoin Transactions.**, Mongardini, A. M., Friolo D., and Ateniese, G. In 21st International Conference on Applied Cryptography and Network Security (ACNS 2023), New York, USA (to appear).
18. **MARTSIA: Enabling Data Confidentiality for Blockchain-based Process Execution**, Marangone E., Di Ciccio C., Friolo D., Nemmi E. N., Venturi D., Weber I., in Proceedings of the 27th International EDOC Conference (EDOC 2023).
19. **On the Complete Non-Malleability of the Fujisaki-Okamoto Transform**, Friolo D., Salvino M., Venturi D. in Proceeding of the 21st International Conference on Applied Cryptography and Network Security (ACNS 2023), Kyoto, Japan.
20. **Compact proofs of partial knowledge for overlapping CNF formulae** Avitabile, G., Botta, V., Friolo, D., Venturi, D., Visconti, I. (2025). In Journal of Cryptology.

Selected Talks

- 2026 **Stealth and beyond: attribute-driven accountability in Bitcoin transactions**
Presented at Applied Cryptography and Network Security (ACNS 2026).
- 2025 **Compact Proofs of Partial Knowledge for Overlapping CNF Formulae**
Presented at ArticCrypt 2025, Longyearbyen, Norway.
- Private Electronic Payments with Self-Custody and Zero-Knowledge Verified Reissuance**
Presented at Financial Cryptography and Data Security Conference Workshops (Co-DecFin), 2025 at Miyakojima, Japan, and, as an invited talk, at the CrypTO Local Conference in Turin, Italy.
- 2024 **Data Redaction in Smart-Contract-Enabled Blockchains**
Presented at the 6th Distributed Ledger Technologies Workshop (DLT) 2024 in Turin, Italy.
- 2023 **Registered (Inner-Product) Functional Encryption**
Presented at the International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT) 2023 in Guangzhou, China.
- 2022 **Shielded Computations in Smart Contracts Overcoming Forks**
Presented at Financial Cryptography and Data Security 2021, Virtual
- 2021 **Terrorist Attacks for Fake Exposure Notifications in Contact Tracing Systems**
Presented at the 19th International Conference on Applied Cryptography and Network Security 2021, Virtual.
- 2019 **On Financial Fairness**
Weekly crypto group talk at CS Dept. Aarhus University, Invited talk at Sapienza University of Rome - CS Dept. (De Cifris Schola Latina seminars)
- A Black-Box Construction of Fully-Simulatable, Round-Optimal Oblivious Transfer from Strongly Uniform Key Agreement**
Weekly crypto group talk at CS Dept. Aarhus University (DK). Presented at Theory of Cryptography Conference in Nuremberg (Dec 2019)
- The Rush Dilemma: Attacking and Repairing Smart Contracts on Forking Blockchains**
Invited talk at Chalmers University (SWE), Lund University (SWE) and Weekly COBRA Seminar at CS Dept. Aarhus University (DK)

Teaching

2026 - June	PhD School Speaker - 4 hours	7th Scientific School on Blockchain & Distributed Ledger Technologies Basic Cryptography for Blockchains
2025/26	Lecturer - 6 CFU (60 hours)	Sapienza University of Rome - MSc in CS/Cybersecurity Security in Software Applications
2025/26	Lecturer - 3 CFU (30 hours)	Sapienza University of Rome - BSc in Computer Science Fondamenti di Programmazione (Programming I)
2024/25	Lecturer - 20 hours	Sapienza University of Rome - PhD course in Cybersecurity Post-Quantum Cryptography Topics: Lattices, Post-Quantum NIST Standards, Basic and Advanced Cryptographic primitives from Lattice-Based assumptions.
2024/25	Lecturer - 6 CFU	Sapienza University of Rome - MSc in Cybersecurity Ethical Hacking
2024/25	Lecturer - 3 CFU	Sapienza University of Rome - BSc in Computer Science Fondamenti di Programmazione (Programming I)
2024/25	Lecturer - 6 CFU	Sapienza University of Rome - MSc in CS/Cybersecurity Security in Software Applications
2023/24	Lecturer - 6 CFU	Sapienza University of Rome - MSc in Cybersecurity Ethical Hacking
2023/24	Lecturer - 3 CFU	Sapienza University of Rome - MSc in CS/Cybersecurity Security in Software Applications
2022/23	Lecturer - 6 CFU	Sapienza University of Rome - MSc in CS/Cybersecurity Security in Software Applications
2021/22	Lecturer - 6 CFU	University of Trento - MSc in Computer Science Cryptography, Complexity and Financial Technologies
2019/20	Teaching assistant	Sapienza University of Rome - BSc in Computer Science Architetture degli Elaboratori, Metodologie di Programmazione

PhD Advising

2023–now	Andrea Reale, PhD Student	Sapienza University of Rome - PhD Course in Cybersecurity. Co-advisor of the PhD Student on Subversion-Resilient Cryptography.
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MSc Thesis Advising

2025	Adele Mussari Thesis Title: Integrating permissioned blockchains with Flutter for private and verifiable distributed storage	Sapienza University of Rome - MSc in Cybersecurity.
2025	Filippo Trotter Thesis Title: Quantum Vulnerabilities in Verifiable Credentials: SD-JWT vs. BBS+ and the Path Forward.	University of Trento - MSc in Math.
2024	Giuseppe Di Naso Thesis Title: Embedding Post-quantum Cryptography inside SmartCards.	Sapienza University of Rome - MSc in Cybersecurity.
2024	Cosmo Greco Thesis Title: Enhancing security by design processes: analysis of best practices, frameworks, technologies and improvement with digital forensics.	Sapienza University of Rome - MSc in Cybersecurity.
2024	Francesco Alessandrini Thesis Title: A Comparative Analysis of Modern Smart Contract Vulnerabilities Detection Mechanisms.	Sapienza University of Rome - MSc in Cybersecurity.
2021	Matteo Castagna Thesis Title: SmartDECO: Fully Decentralized Markets Based On Ethereum Smart Contracts And Decentralized Tls Oracles.	University of Trento - MSc in Computer Science.
2021	Helidona Shabani Thesis Title: Blockchain Securitization: An Innovative And Decentralized Technology To Tokenize And Implement Digital Assets.	University of Trento - MSc in Computer Science.

Committees and Boards

2027	Financial Cryptography 2027 Program Committee Member	Barbados
2027	Financial Cryptography 2026 Contact Editor	St. Kitts
2026	10th Workshop on Trusted Smart Contracts (WTSC) 2026 (co-located with Financial Cryptography and Data Security 2026) Program Chair	St. Kitts
2026	10th Workshop on Trusted Smart Contracts (WTSC) 2026 (co-located with Financial Cryptography and Data Security 2026) Program Committee Member	St. Kitts
2026	IACR PKC 2026 Program Committee Member	West Palm Beach, USA
2026	IACR EUROCRYPT 2026 Workshop Chair	Rome, Italy
2025–ongoing	IEEE Transactions on Information Forensics and Security Consulting Area Editor	
2024	AI2Sec2024 Workshop (co-located w. CCS) Program Committee Member	Salt Lake City, USA
	CIFRIS 2024 Program Committee Member	Rome, Italy
2023	CIFRIS 2023 Conference co-Chair and Program Committee Member	Rome, Italy
2023	IACR ASIACRYPT 2023 Session Chair	Guangzhou, China
2022	ACNS 2022 Session Chair	Rome, Italy

Other Experiences and Memberships

2024–now	Member of Istituto di Alta Matematica (INdAM) of Gruppo Nazionale per il Calcolo Scientifico Fondamenti dell'Informatica e sviluppo di Sistemi Software
2025	Where: Università Roma Tre, Roma Jury of the IXH25 XRP Cryptography Hackaton
2024	Where: Ministero Degli Esteri, Rome Invited to join the conference “Ecosistema Nazionale di Cyber Capacity Building” as a representative of Dipartimento di Informatica, Sapienza University of Rome, Rome, Italy.
2024	Where: Sala del Refettorio, Camera dei deputati, Rome Invited to join the conference “Crittografia e imprese per il Paese” as a researcher and member of DeCifris association.

Peer Review

2025	IEEE Symposium on Security and Privacy (IEEE S&P), the 44th Annual International Cryptology Conference (CRYPTO), Transactions on Information Forensics and Security (TIFS), IEEE Transactions on Dependable and Secure Computing (TDSC), 35th USENIX Security Symposium.
2024	14th Conference on Security and Cryptography for Networks (SCN), International Colloquium on Automata, Languages, and Programming (ICALP), IEEE International Conference on Web Services (ICWS), Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT), ACM Conference on Computer and Communications Security (CCS), International Conference on Practice and Theory in Public Key Cryptography (PKC), Transactions on Information Forensics and Security (TIFS)
2023	International Conference on Applied Cryptography and Network Security (ACNS), European Symposium on Research in Computer Security (ESORICS), Elsevier Theoretical Computer Science (TCS), Elsevier Vehicular Communications
2022	Advances in Cryptology (EUROCRYPT), 13th Conference on Security and Cryptography for Networks (SCN), European Symposium on Research in Computer Security (ESORICS), Annual International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT)
2020	Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT), International Conference on Applied Cryptography and Network Security (ACNS), 19th International Conference on Cryptology and Network Security (CANS), Annual International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT)
2019	International Cryptology Conference (CRYPTO), IEEE Transactions on Information Forensics and Security (TIFS), International Conference on Applied Cryptography and Network Security (ACNS)
2018	IEEE Transactions on Information Forensics and Security (TIFS), International Conference on Practice and Theory in Public Key Cryptography (PKC)

Autorizzo il trattamento dei miei dati personali presenti nel curriculum vitae ai sensi del Decreto Legislativo 30 giugno 2003, n. 196 e del GDPR (Regolamento UE 2016/679).